M1 Oscillator

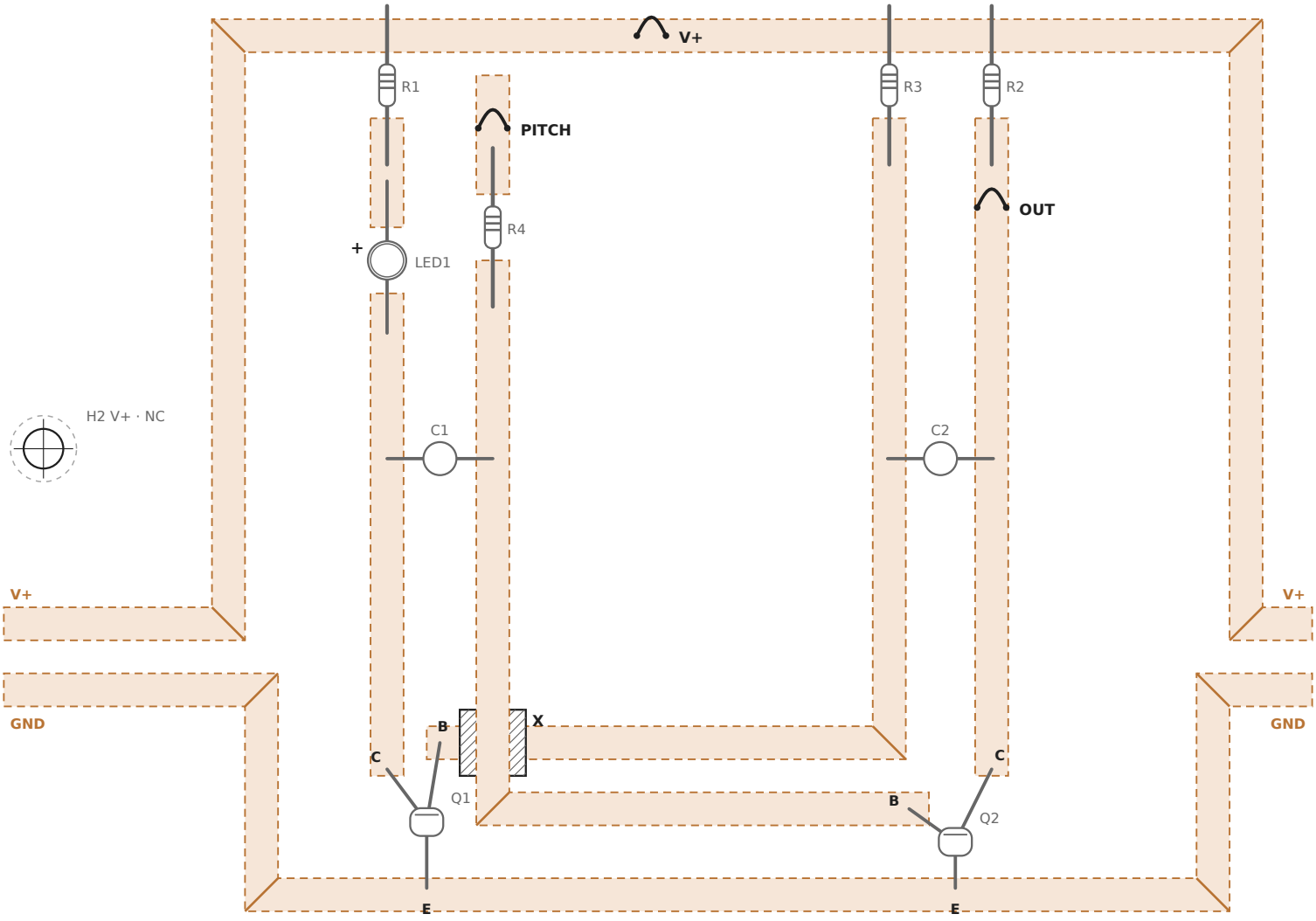
Two transistors switch each other on and off: that's a tone.

Draft: tagged steps may change

H1 BUS A · NC

You need

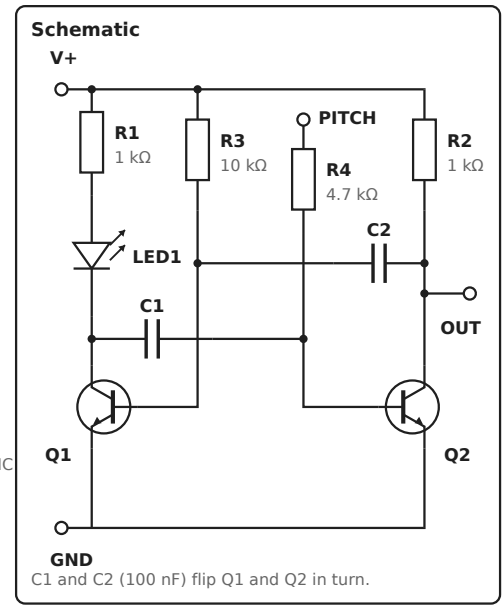
- Q1, Q2: 2N3904 transistors
- C1, C2: 100 nF capacitors, no polarity
- R1, R2 1 kΩ · R3 10 kΩ · R4 4.7 kΩ
- LED1: red, 5 mm
- Bare wire for 3 patch loops; 22 kΩ to test
- Copper tape, solder, 2-core cable



H2 V+ · NC

H3 GND · NC

Patch PITCH to a controller such as M2's keyboard, and OUT to an IN on M7.



H4 BUS B · NC

Build it

1. Lay the power ring, then the seven lanes inside it, one piece of tape each. Fold at the creases.
2. Where two lanes cross, lay the lower one, stick the insulating patch over it, then lay the upper one.
3. Place R1 to R4, C1 and C2 across their gaps, either way round. Place LED1 as printed: long leg on the +.
4. Lay Q1 and Q2 flat face up and bend each leg to its letter: E, B and C.
5. Solder every lead to its lane, or hold it flat with a small patch of copper tape. **[T0]**
6. Solder a patch loop upright on each loop mark: V+, PITCH and OUT. **[T0]**
7. Join the edge connectors to the pages either side with 2-core cable, or with red and black clip leads.

Play it

- To test, clip a 22 kΩ resistor between the V+ and PITCH loops: M7 should play a steady note, and LED1 glows.
- Then patch PITCH to M2's stylus. More resistance on PITCH gives a lower note.

[T0] and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.

Print at 100 %: this bar should measure 100 mm.